PAPER_936: GW170817 Inspiral Phase Lag

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 212 **Source:** ns_phonon_gw170817_wstp.py (GW170817InspiralPhaseLag) **Calculator:** GW170817InspiralPhaseLagCalc (CP4 #520) **CVW:** v2.0.0 compliant

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Abstract

We calculate the cumulative phase lag induced by phonon suppression during the GW170817 inspiral, from gravitational wave frequency f_0 = 20 Hz to f_max = 300 Hz. The UQFF predicts a total phase difference Delta_Phi ~ 2310.8 rad (367.8 cycles) between the phonon-suppressed waveform and the pure GR template. This phase lag arises from the frequency-dependent phonon damping D_total = 0.333 modulated by the 26-layer suppression sum.

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1. Core Equations

Accumulated phase lag:

$$\Delta\Phi = 2\pi(f_{\text{max}} - f_0) \cdot D_{\text{total}} \cdot \frac{\Phi}{\Phi_0}$$

where: - $f_0 = 20$ Hz (LIGO low-frequency cutoff) - $f_{\text{max}} = 300$ Hz (approximate merger frequency) - $D_{\text{total}} = 0.333$ - $\Phi/\Phi_0 = S_{26}$ at resonance

In cycles:

$$\Delta\Phi_{\text{cycles}} = \frac{\Delta\Phi}{2\pi} \approx 367.8 \text{ cycles}$$

Key Parameters

Parameter	Value
M_chirp	1.188 M_sun
f_0	20 Hz
f_max	300 Hz
D_total	0.333
Delta_Phi	2310.8 rad
Delta_Phi_cycles	367.8

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2. UQFF Integration

The `GW170817InspiralPhaseLagCalc` (CP4 #520) computes Delta_Phi as a function of D_total with simulate() sweeping D_total = [0.2, 0.333, 0.5, 0.7]. This maps the sensitivity of the phase lag to the overall phonon suppression strength.

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3. Physical Significance

Phase-coherent matched filtering is the primary detection technique for compact binary inspirals. A phase lag of ~368 cycles would be detectable in current LIGO data as a systematic template mismatch. However, this lag is partially degenerate with mass and spin parameters. Breaking this degeneracy requires multi-messenger observations (electromagnetic counterpart timing) or next-generation detector networks with improved low-frequency sensitivity.

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4. Source Data

- **File:** ns_phonon_gw170817_wstp.py
- **Session:** 212
- **CP4 Class:** GW170817InspiralPhaseLagCalc (#520)

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Abbott, B.P. et al. (LIGO/Virgo) – GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral, PRL 119, 161101 (2017)
3. Cutler, C. & Flanagan, E.E. – Gravitational waves from merging compact binaries, PRD 49, 2658 (1994)